

Jump Process

(1. Stochastics with jumps)

Why jumps ?

Three main reasons why jumps improve capturing real-world dynamics

$t \xrightarrow{\text{cont.}} S_t$

I) **Black-Scholes** assumes stock price evolves continuously i.e.; price cannot jump suddenly. But real markets often exhibit sudden jumps which cannot be explained by Brownian shocks. Large price moves can occur in a short time due to defaults, crashes, geopolitical shocks, central bank policies etc....

$(w), dw \sim N(0, dt)$

II) Stock returns often exhibit fat tails (leptokurtic) compared to the normal distribution assumption as enforced by Brownian motion. Jumps can increase the probability of extreme outcomes.

III) Jumps help explain volatility smile/skew.

Bsm 1973
Merton 1976

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t \quad \rightarrow \quad \frac{dS_t}{S_t} = \mu dt + \sigma dW_t + \underbrace{(Y - 1)dN_t}_{(\text{jump})}$$



Recap Poisson Process

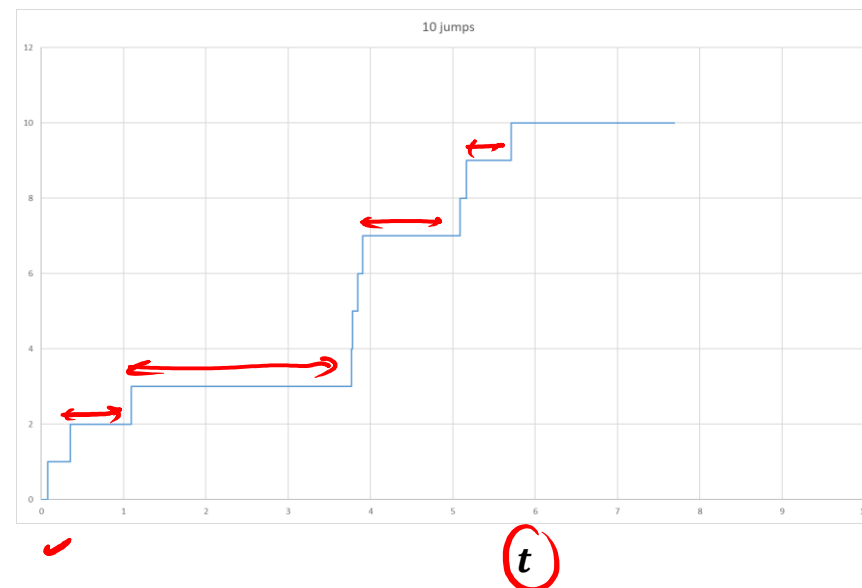
The most elementary jump process is a Poisson process (N_t) which is a counting process with the following properties

✓ 1. $N_0 = 0$

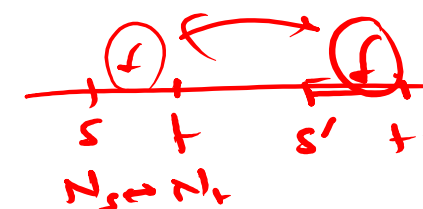
✓ 2. $\mathbb{P}(N_t - N_s = k) = e^{-\lambda(t-s)} \frac{(\lambda(t-s))^k}{k!}$

(increments $N_t - N_s$ are independent and follow Poisson distribution with parameter $\lambda(t-s)$)

Each arrival (jump) is of size 1. The time between two arrivals follows exponential distribution with parameter λ



$\lambda = 5 / \text{hr}$
 $(\text{day}) = 120 \text{ arrivals}$
 $\lambda_{\text{day}} = 120 / \text{day}$
 $P(X=120) = \frac{e^{-120} \times (120)^{120}}{120!}$



Compensated Poisson Process

What is the mean and variance of the counting process N_t ?

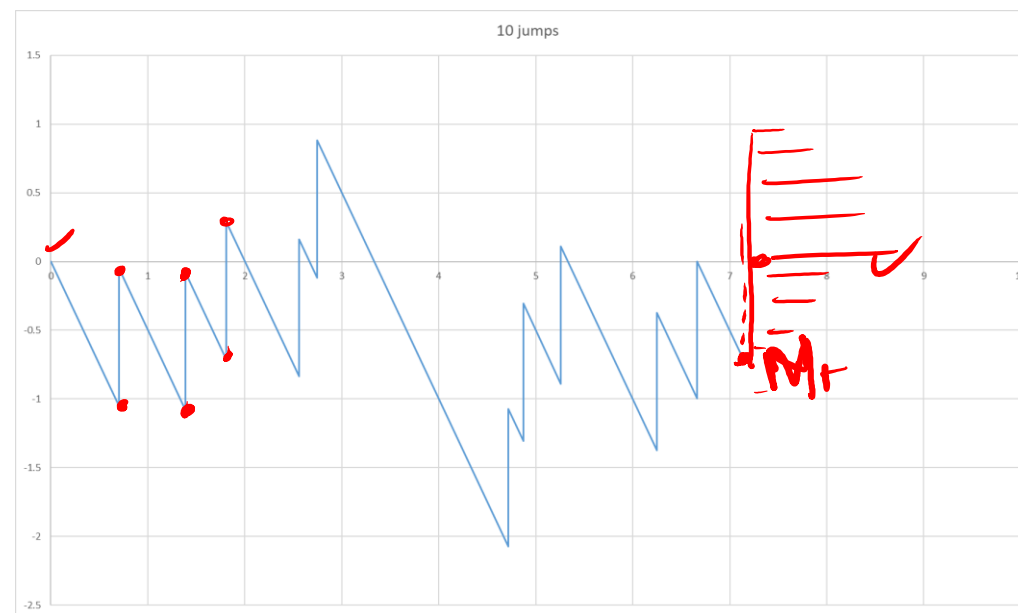
$\rightarrow \mathbb{E}[N_t] = \lambda t$
 $\rightarrow \mathbb{V}[N_t] = \lambda t$

$N_t = N_t - N_0 \sim \text{Poi}(\lambda t)$
 λt

The counting process has a drift of λt . We define the Compensated Poisson process as $M_t = N_t - \lambda t$. Note that M_t is a martingale.

$\mathbb{E}[M_t] = \mathbb{E}[N_t] - \lambda t = 0$

$-\lambda t$



Compound Poisson Process

A plain counting process does not give much flexibility as the jump size = 1. We would be better off if we could model the size of each jump as well. Let's consider a sequence of $\{Z\}_i$ as a sequence of i.i.d random variables which denote the size of the jump. The Compound Poisson process is defined as the sum of a random number of such jumps.

$$\mu dt + \sigma dw \sim N(0, dt)$$

$$Y_t = Z_1 + Z_2 + \dots + Z_{N(t)} = \sum_{i=1}^{N(t)} Z_i$$

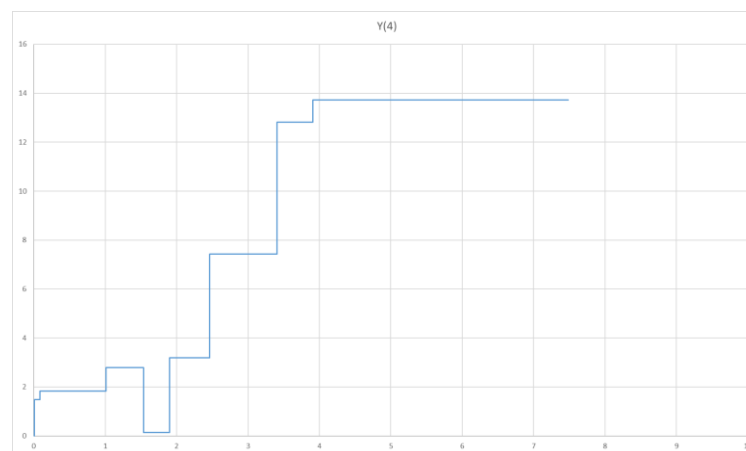
$Z_i \sim \text{random}$ (handwritten note: $Z_i \sim$ random)

$N(t)$ (handwritten note: how many jumps)

$Z_i \sim$ (handwritten note: $Z_i \sim$)

$\mathbb{E}(Z_i) = \mu$ (handwritten note: $\mathbb{E}(Z_i) = \mu$)

- 1) Show that the mean of Y_t is $\mathbb{E}[Y_t] = \lambda t \mathbb{E}[Z]$ (Hint : Use Tower property of iterative expectation)
- 2) What would be the expression for compensated compound Poisson process ?



Cadlag (RCLL)

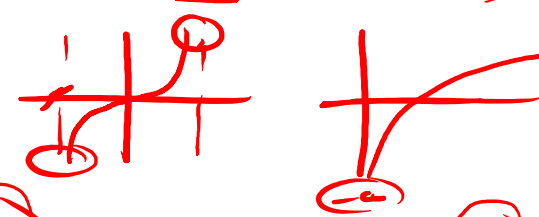
In mathematics, a cadlag or RCLL (right-continuous with left limits) is a function defined in $\mathbb{R}$ which is right-continuous everywhere and left limit exists everywhere.

left limit exists

$$f(t^-) := \lim_{s \rightarrow t^-} f(s) \text{ exists}$$

right limits exists

$$f(t^+) := \lim_{s \rightarrow t^+} f(s) \text{ exists and is equal to } f(t)$$

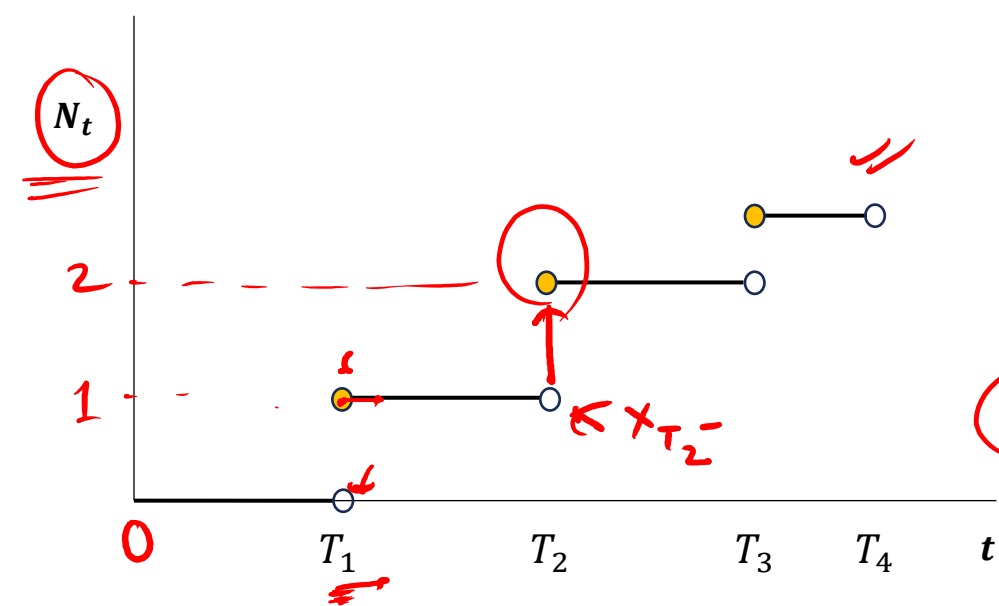
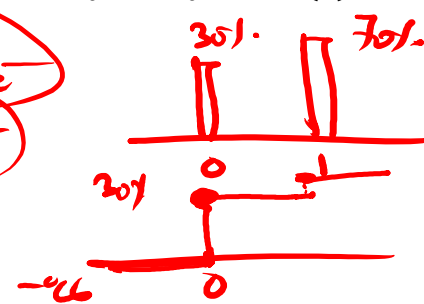


$$\begin{aligned} \rightarrow N_{T_1^-} &= 0 \\ \rightarrow N_{T_1} &= 1 \\ \rightarrow N_{T_1^+} &= 1 \end{aligned}$$

Note : the jump gets added to the t^- instant as t already includes the jump

$$N_t = N_{t^-} + Z(t)$$

discrete CDF



Infinitesimal time behavior

How does the distribution look like in infinitesimal time (dt) ?

$$\begin{aligned}
 \mathbb{P}(N_{t+dt} - N_t = 1) &= e^{-\lambda dt} \frac{(\lambda dt)^1}{1!} = \lambda dt + o(dt) \\
 \mathbb{P}(N_{t+dt} - N_t = 0) &= e^{-\lambda dt} \frac{(\lambda dt)^0}{0!} = 1 - \lambda dt + o(dt) \\
 \mathbb{P}(N_{t+dt} - N_t > 1) &= o(dt)
 \end{aligned}$$

Handwritten notes:

- dN (circled)
- $dt \leftarrow SDE$ (circled)
- λdt (circled)
- $t \rightarrow t + dt$
- $dt \rightarrow 0$
- $N_t \rightarrow N_{t+dt}$
- dN (circled)

Handwritten notes:

- $dN = 0$ ✓
- $= 1$ ✓
- Matrix: $\begin{bmatrix} 2 & 1 \\ 3 & i \\ 6 & j \end{bmatrix}$

Essentially the Poisson increment in infinitesimal period behaves like a Bernoulli random

Handwritten notes:

- $e^{-\lambda dt} \cdot (\lambda dt)$
- $\approx (1 - \lambda dt + \dots) \lambda dt$
- $= \lambda dt -$

Handwritten notes:

- dt (circled)
- $dN_t = \begin{cases} 1, & \text{prob} = \lambda dt \\ 0, & \text{prob} = 1 - \lambda dt \end{cases}$

Ito's Lemma for Poisson Process – (I)

Just like we performed Ito calculus on Brownian motion and functions on Brownian motion, we need to develop the machinery for jump process. Consider $N(t)$ as our counting process with intensity λ . How do we develop the SDE for a function that depends on $N(t)$?

$\left(\frac{ds}{dN}\right)$

$\xrightarrow{dt}$
 $t \rightarrow t + dt$

$dN_t = \begin{cases} 1, & \text{prob} = \lambda dt \\ 0, & \text{prob} = 1 - \lambda dt \end{cases}$

$df(N_t) = ?$

$$\begin{aligned} df(N_t) &= f(N_{t+dt}) - f(N_t) \\ &= f(N_t + dN_t) - f(N_t) \\ &= [f(N_t + 1)dN_t + f(N_t + 0)(1 - dN_t)] - f(N_t) \\ &= [f(N_t + 1) - f(N_t)]dN_t \end{aligned}$$

$0 \rightarrow df = 0$
 $1 \rightarrow df = f(N_t + 1) - f(N_t)$

$\left(\frac{df}{dN}\right)$

$df(N_t) = [f(N_t + 1) - f(N_t)]dN_t$

SDE

Note that it's a full revaluation as the jump is of finite size (not infinitesimal)



Ito's Lemma for Poisson Process – (II)

Now consider a function that depends on both t and N_t

$$t \rightarrow t + dt$$

$$dN_t = \begin{cases} 1, & \text{prob} = \lambda dt \\ 0, & \text{prob} = 1 - \lambda dt \end{cases}$$

$$\underline{df(t, N_t)} = ?$$

$$df = f(\underline{t+dt}, \underline{N_t+dN_t}) - f(\underline{t}, \underline{N_t})$$

$$df(t, N_t) = \left(\frac{\partial f(t, N_t)}{\partial t} dt \right) + \left[f(t, N_t + 1) - f(t, N_t) \right] dN_t$$

The change comes from two sources; explicit time dependence (continuous drift) + probabilistic 1 jump



Ito's Lemma for Poisson Process – (III)

Now consider a more general stochastic process X_t and a function $f(X_t)$ with dynamics

$$ds/s = \mu dt + \sigma dw$$

$$ds = \mu s dt + \sigma s dw$$

$$\text{SDE } dX_t = \mu(t, X_t)dt + v(t, X_t)dN_t$$

$$S \rightarrow W_t \text{ (GBM)}$$

$$V(t, S)$$

$$df(t, X_t) = \left[\frac{\partial f(t, X_t)}{\partial t} + \frac{\partial f(t, X_t)}{\partial X_t} \mu(t, X_t) \right] dt + [f(t, X_{t-} + v(t, X_t)) - f(t, X_t)] dN_t$$

It's very important to note the use of time instant in the above equation. The jump always needs to be added to X_{t-} as X_t already contains the jump (RCLL).

$$X_{t-} + v(t, X_{t-})$$

$$(X_t) + v(t, X_t)$$



Ito's Lemma for Jump Diffusion models – (IV)

We can also add a diffusion term



$$dX_t = \underbrace{\mu(t, X_t)dt}_{\text{drift}} + \underbrace{\sigma(t, X_t)dW_t}_{\text{diffusion}} + \underbrace{v(t, X_t)dN_t}_{\text{jump}} \quad \leftarrow \text{SDE} \quad \text{Median}$$

$$df(t, X_t) = \underbrace{\left[\frac{\partial f(t, X_t)}{\partial t} + \frac{\partial f(t, X_t)}{\partial X_t} \mu(t, X_t) + \frac{1}{2} \sigma^2(t, X_t) \frac{\partial^2 f(t, X_t)}{\partial X_t^2} \right] dt + \sigma(t, X_t) \frac{\partial f(t, X_t)}{\partial X_t} dW_t}_{\text{(continuous part)}} + \underbrace{[f(t, X_{t-} + v(t, X_t)) - f(t, X_t)] dN_t}_{\text{(discrete jump impact)}}$$



Ito's table

Now that we have a new source of randomness, we want to enhance the Ito table for product rule.

$$dN = \begin{cases} 1, & \lambda dt \\ 0, & 1 - \lambda dt \end{cases}$$

$$(dN)^2 = \begin{cases} 1, & \lambda dt \\ 0, & 1 - \lambda dt \end{cases}$$

	dt	dW ^{$\sqrt{dt}$}	dN
dt	0 ✓	0 ✓	0 ✓
dW	0	dt	0
dN	0	0	dN

$$dN \cdot dt = \begin{cases} dt, & \lambda dt \\ 0, & 1 - \lambda dt \end{cases}$$

$$\mathbb{E}[dN \cdot dt] = \lambda (dt)^2 \rightarrow 0$$

$$\mathbb{V}(dN \cdot dt) = (dt - 0)^2 \cdot \lambda dt + (0 - 0)^2 \cdot (1 - \lambda dt)$$

$$= \lambda dt^3 \rightarrow 0$$

1) Prove $dN^2 = dN$ ✓

2) If $dX_t = adt + bdW_t + cdN_t$ & $dY_t = edt + fdW_t + gdN_t$, then what is $dX_t \cdot dY_t = ?$

$$= bf dt + cg dN$$



Exercises



1. Two assets can jump with intensities λ_1 and λ_2 , respectively.
 - a) What is the probability that asset 1 jumps first ? ←
 - b) What is the probability that we observe 2 jumps from asset 1 before 2 jumps from asset 2 ?

